

Implicit Functions; Related Rates

MATH 145 – Applied Calculus

Torin Quinlivan

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Implicitly-Defined Functions

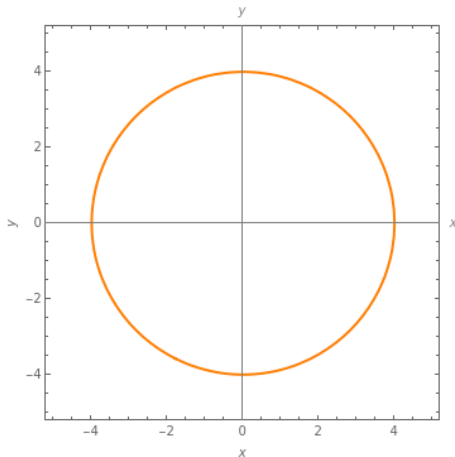
So far, we've worked solely with functions that can be described explicitly in terms of x , i.e. $f(x) = x^3 + 2x^2 + 3x + 4$.

But not all curves can be described this way.

Examples of Implicitly-Defined Functions

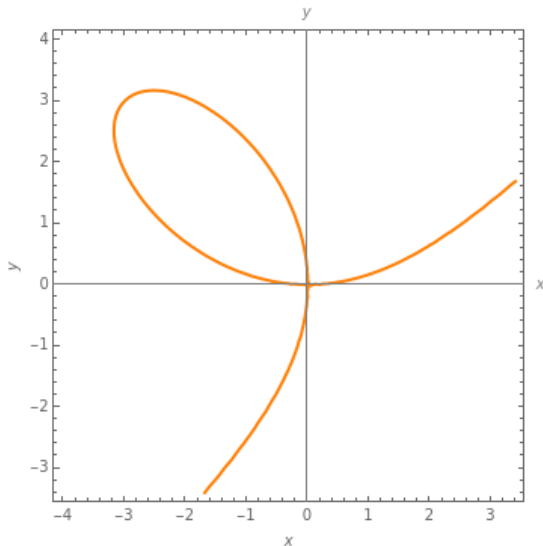
The most obvious example is a circle.

Here is the plot of $x^2 + y^2 = 16$, the circle centered at the origin with radius 4:



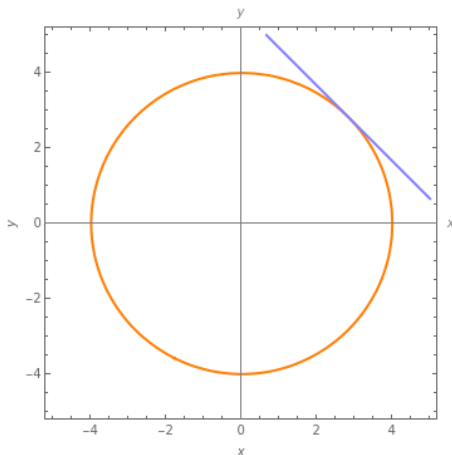
Examples of Implicitly-Defined Functions

Here is the plot of $x^3 - y^3 = 6xy$:



Examples of Implicitly-Defined Functions

What if we want to find the rate of change of the function $x^2 + y^2 = 16$ at the point $(2\sqrt{2}, 2\sqrt{2})$?



Implicit Differentiation

Consider the circle $x^2 + y^2 = 16$.

We can't differentiate the normal way, but what if we think of y as a *implicit function of x* , but which we can differentiate?

Now we can't find the derivative all the way, we won't know the derivative, $\frac{dy}{dx}$, directly (yet). But we can use the Chain Rule!

Example #1: Implicit Differentiation

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} [16]$$

$$\frac{d}{dx} [x^2] + \frac{d}{dx} [y^2] = 0$$

$$2x + 2y \frac{dy}{dx} = 0$$

As by the Chain Rule:

$$\frac{d}{dx} [y(x)^2] = 2(y(x))^1 \frac{dx}{dy}$$

Example #1: Implicit Differentiation

From here, solve for $\frac{dy}{dx}$ to find the form of the derivative:

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

What is the slope of the tangent line at $(2, 2\sqrt{3})$?

Example #2: Implicit Differentiation

1. Find the slope of the tangent line of the curve $y^3 + x^3 - 3xy = 0$ at the point $\left(\frac{3}{2}, \frac{3}{2}\right)$.

Example #3: Implicit Differentiation

- Find a formula for $\frac{dy}{dx}$ for the curve $x = y^5 - 5y^3 + 4y$.
- Where does the graph $x = y^5 - 5y^3 + 4y$ have a vertical tangent line?

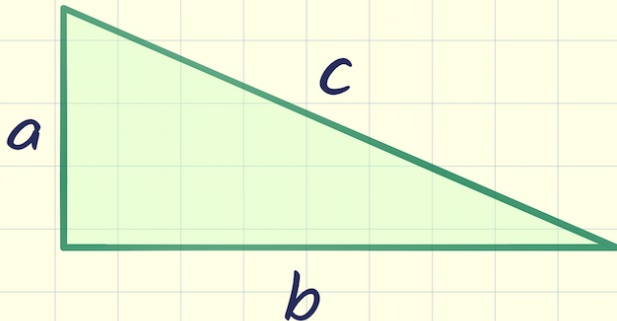
Example #4: Implicit Differentiation

Consider the curve given by the equation $2y^3 + y^2 - y^5 = x^4 - 2x^3 + x^2$.

4. Find the formula for $\frac{dy}{dx}$.
5. Where is the tangent line horizontal?
6. Where is the tangent line vertical?

Related Rates

Consider a right triangle with the three sides changing. How can we relate the rates of change of the lengths of the three sides?



Related Rates

The three sides are related through the Pythagorean Theorem
 $a^2 + b^2 = c^2$.

Each of the lengths of the three sides of the triangle are changing as a function of time, but are also tied to each other.

Related Rates

So what if we derive all three in terms of the time, t ?

$$\begin{aligned}\frac{d}{dt} [a^2 + b^2] &= \frac{d}{dt} [c^2] \\ \frac{d}{dt} [a^2] + \frac{d}{dt} [b^2] &= \frac{d}{dt} [c^2] \\ 2a \frac{da}{dt} + 2b \frac{db}{dt} &= 2c \frac{dc}{dt}\end{aligned}$$

Now if we know the rates of change of two of the sides, we can tell the rate of change of the third definitively.

Example #5: Related Rates

Water is draining from the bottom of a cone-shaped funnel at a rate of $0.03 \text{ ft}^3/\text{sec}$. The height of the funnel is 2 feet and the radius at the top of the funnel is 1 foot. At what rate is the height of the water in the funnel changing when the height of the water is $1/2$ feet?